

7.2 The sources, origins, physical and working properties of each natural and manufactured timber and their social and ecological footprint	To apply knowledge and understanding of the advantages, disadvantages and applications of the following materials, in order to be able to discriminate between them and select appropriately.
	7.2.1 Natural timbers – hardwoods: a oak (in topic 1) b mahogany (in topic 1) c beech (in topic 1) d balsa (in topic 1) e jelutong f birch g ash.
	7.2.2 Natural timbers – softwoods: a pine (in topic 1) b cedar (in topic 1) c larch.
	7.2.3 Manufactured timber: a plywood (in topic 1) b medium density fibreboard (MDF) (in topic 1) c chipboard.
	7.2.4 Sources and origins – where natural and manufactured timbers are resourced/manufactured and their geographical origin: a Alpine forests – pine, cedar, larch b European forests – oak, beech, ash, birch c Amazonian forests – mahogany.
	7.2.5 The physical characteristics of each timber: a knots b colour c grain structure d density.

TIMBERS

Key idea	What students need to learn:
	7.2.6 Working properties – the way in which each material behaves or responds to external sources: - a hardness (in topic 1) - b toughness (in topic 1) - c durability (in topic 1) - d elasticity - e tensile strength - f compressive strength.
	7.2.7 Social footprint: - a trend forecasting - b impact of logging on communities - c ease and difficulty of recycling and disposal.
	7.2.8 Ecological footprint: - a sustainability - b deforestation - c habitat destruction and loss - d processing - e transportation - f wastage - g pollution.